

Advanced Trigonometry

Self-Test 2

Allow 25 minutes to complete this test.

1. Find the length of side a in $\triangle ABC$ if $A = \frac{\pi}{3}$, and $b = 6$ and $c = 8$.
2. Find the angle A in $\triangle ABC$ if $b = 4$, $c = 8$ and $B = \frac{\pi}{6}$.
3. A 10 metre pole is leaning toward the sun. If the angle of elevation of the sun is 32° and the shadow of the pole is 12 metres, what angle does the pole make with the horizontal?
4. Graph the function $y = f(x) = \frac{1}{3} \cos \frac{x}{2}$ noting its amplitude and period.
5. Graph the function $y = f(x) = 4 \tan 2x$.
6. If $\sec x = -\frac{3}{2}$ and $\frac{\pi}{2} \leq x \leq \pi$, evaluate $\cot x$ using trigonometric identities.
7. Simplify the expression $\frac{\sin x \sec x}{\tan x + \cot x}$.
8. True or false: $\cos\left(\frac{3\pi}{2} - x\right) = \sin x$.
9. True or false: $\sin(x + y) + \sin(x - y) = 2 \sin x \cos y$.
10. Show that $(\sec x - \tan x)^2 = \frac{1 - \sin x}{1 + \sin x}$.